Field Hours

# CSCE 704 Data Analytics for Cybersecurity

Original notes for this site. Same topics as the catalog course. Not a reprint of another university's lectures, and not academic credit.

Lecture 1 · 90 minutes

## A row you can sort twice

You start with a row, not with a model. The mechanism is a fixed set of columns you can fill from a log. This toy row has 5 fields: time, source address, account, action, and result. A free-text paragraph is not a row, because you cannot sort it the same way twice. You drop or repair a row that is missing the account when the question is about accounts.

You separate the label from the features. The label is whether the row is malicious, and it must come from a decision you can cite, not from the same field you will predict. If you mark a row malicious because the action string contains the word fail, you have leaked the label into a feature. Suppose you have 40 rows and you labeled 6 by hand from a ticket. The other 34 stay unlabeled until you have a reason.

Five fields, and a label that is not one of them.

Time of the event

Source address

Account

Action and result

Label from a ticket

### Six labels, thirty-four waiting

1. You have 40 rows with the 5 fields filled. A ticket supports labels on 6 of them.

2. The other 34 stay unlabeled. You do not mark a row malicious because the action text says fail.

3. A row missing the account is dropped or repaired before any count about accounts. Free text never becomes a sixth hidden column.

Problem 1. You have 40 rows and tickets support labels on 6. How many rows stay unlabeled?

Answer. 34. 40 minus 6 is 34. Those 34 stay unlabeled until you have a reason. Copying the word fail from the action field into the label is leakage, not a reason.

Problem 2. The question is about accounts, and a row has time, address, action, and result, but no account. What do you do with it before you count?

Answer. You drop it or repair it. The toy row has 5 fields, and account is one of them. A paragraph of free text is not a substitute column.

Lecture 2 · 90 minutes

# The denominator is the story

A base rate is the count of the class divided by the count of rows, and it is the mechanism behind every later percentage. If 2 of 50 flows are malicious, the base rate is 2/50, which is 1/25, or 0.04. A detector that flags everything scores a recall of 1 and is still useless, because 48 of the 50 flags are false. You write the base rate on the page before you praise a model.

You keep the units straight when you compare two days. Day 1 has 2 malicious flows out of 50, rate 0.04. Day 2 has 2 malicious flows out of 200, rate 0.01. The count stayed at 2, and the rate fell by a factor of 4, because 0.04 divided by 0.01 is 4. You do not say the attack doubled. You say the count was flat and the denominator changed.

Same count, different rate.

Day 1

2 malicious out of 50

Rate is 0.04

Write this before the model

Day 2

2 malicious out of 200

Rate is 0.01

The count did not double

## Two malicious, two denominators

1. Day 1: 2 of 50, which is 0.04. Day 2: 2 of 200, which is 0.01.

2. The ratio of the rates is 0.04 divided by 0.01, which is 4. The malicious count stayed 2.

3. A detector that flags all 50 flows on day 1 has 48 false flags. You write 0.04 on the page before you talk about that detector.

Problem 1. Day 1 has 2 malicious flows out of 50, and day 2 has 2 out of 200. What are the two rates, and by what factor did the rate fall?

Answer. The rates are 0.04 and 0.01. The factor is 4, because 0.04 divided by 0.01 is 4. The count stayed 2, so you do not say the count doubled.

Problem 2. A detector flags all 50 flows, and 2 are malicious. How many flags are false?

Answer. 48. 50 minus 2 is 48. Recall can be 1 and the detector is still useless at a base rate of 2/50, which is 0.04. Write that rate before you praise the model.

Lecture 3 · 90 minutes

# Precision is a fraction

A supervised detector maps features to a score, and the mechanism you control is the threshold. You flag a row when its score is at least 7. In a batch of 10 scores, the values 9, 8, and 7 are flags, and the value 6 is not. That is 3 flags. If 2 of those 3 are truly malicious, precision is 2/3. You report 2/3, not a claim that the model works.

You train on rows that already have labels, and you test on rows you did not train on. If you have 40 labeled rows, a 30 and 10 split means you fit on 30 and you measure precision on 10. Measuring on the 30 is not a test. A threshold you picked because it looked good on the test 10 is no longer an honest test of those 10. You would need a fresh slice.

Fit, then measure on rows you held out.

Fit on 30 labeled rows -> Score the other 10 -> Flag scores of at least 7 -> Report precision as a fraction

### Three flags, two true

1. Scores at or above 7 are flags. In the batch, 9, 8, and 7 qualify, and 6 does not. That is 3 flags.

2. Two of the 3 are truly malicious. Precision is 2/3. You write the fraction.

3. Those 3 were inside the held-out 10, not inside the 30 you fit. If you chose the threshold 7 because of these 10, you need a fresh slice before you quote 2/3 again.

Problem 1. The threshold is 7. The scores include 9, 8, 7, and 6. How many of these four are flags, and what is precision if 2 of the flags are true?

Answer. 3 flags. The score 6 is not a flag. Precision is 2/3. You report 2/3, and you do not measure it on the 30 rows you trained on.

Problem 2. You have 40 labeled rows and you fit on 30. How many rows are in the test split?

Answer. 10. 40 minus 30 is 10. Measuring precision on the 30 is not a test. A threshold chosen to look good on that same 10 needs a fresh slice.

Lecture 4 · 90 minutes

## Distance, then a cutoff

Clustering groups rows by distance, with no label. The mechanism is a distance you compute and a cutoff you state. Take two points in the plane, (0, 0) and (3, 4). The squared distance is 3 squared plus 4 squared, which is 9 plus 16, which is 25, so the distance is 5. If your cutoff is 6, these two points can share a cluster. A cutoff of 4 would separate them.

You do not treat cluster size as malice. A cluster of 30 rows can be the nightly backup, and a cluster of 2 can be a mistake in your distance. You name a cluster by the feature that pulled it together, such as the same account on 30 rows. If you change the cutoff from 6 to 2, you should expect the 30 to split, and you recompute before you brief anyone.

A cluster is a cutoff plus a distance.

1. Square each coordinate gap

2. Add, then take the root

3. Compare with the cutoff

4. Name the shared feature

### Three, four, and a cutoff of six

1. The points are (0, 0) and (3, 4). 3 squared is 9 and 4 squared is 16. 9 plus 16 is 25, and the distance is 5.

2. The cutoff is 6, and 5 is within 6, so these two points may share a cluster. A cutoff of 4 would refuse them.

3. If that cluster later has 30 rows, you name the feature they share. You do not brief the 30 as malice, and you recompute if the cutoff moves from 6 to 2.

Problem 1. What is the distance between (0, 0) and (3, 4)? Show the sum of squares.

Answer. 5. 3 squared is 9, 4 squared is 16, and 9 plus 16 is 25. The positive root of 25 is 5. A cutoff of 6 includes the pair. A cutoff of 4 does not.

Problem 2. A cluster contains 30 rows that share an account, and you lower the cutoff from 6 to 2. What must you do before you brief the 30 as one group?

Answer. Recompute the clusters at cutoff 2. The 30 may split. Size is not malice, and a nightly backup of 30 rows is still a backup until the distance says otherwise.

Lecture 5 · 90 minutes

## Far from a quiet mean

Anomaly detection flags a point far from a summary of the rest, and the mechanism is a distance from the mean. Take the four counts 3, 4, 4, and 5. Their sum is 16, so the mean is 16 divided by 4, which is 4. A new count of 10 sits at distance 6 from that mean. If your rule flags a distance of 5 or more, you flag 10, and you do not flag 5, whose distance is 1.

You choose the summary from data you believe is ordinary. If the mean of 4 was computed with the 10 already inside, the mean moves and the distance shrinks, which hides the point you wanted to see. With values 3, 4, 4, 5, and 10, the sum is 26 and the mean is 26 divided by 5, which is 5.2. Distance from 10 to 5.2 is 4.8, and a threshold of 5 would miss it.

The mean you fit changes the distance.

Quiet baseline

Mean of 3, 4, 4, 5 is 4

Distance of 10 is 6

Threshold 5 flags it

Point folded in

Sum 26 over 5 is 5.2

Distance of 10 is 4.8

Threshold 5 misses it

### Sixteen divided by four

1. The quiet counts are 3, 4, 4, and 5. The sum is 16 and the mean is 4.

2. The new count 10 has distance 6. Your rule flags distance 5 or more, so you flag 10. The count 5 has distance 1 and stays quiet.

3. If you instead average 3, 4, 4, 5, and 10, the sum is 26 and the mean is 5.2. The distance 4.8 no longer clears a threshold of 5.

Problem 1. The quiet counts are 3, 4, 4, and 5. What is their mean, and what is the distance of a new count of 10?

Answer. The mean is 4, because the sum 16 divided by 4 is 4. The distance is 6, because 10 minus 4 is 6. A threshold of 5 flags 10. The count 5 has distance 1 and is not flagged.

Problem 2. You fold 10 into the same list, so the values are 3, 4, 4, 5, and 10. What is the mean, and does distance from 10 clear a threshold of 5?

Answer. The mean is 5.2, because the sum is 26 and 26 divided by 5 is 5.2. The distance is 4.8, which does not clear 5. Fit the mean on the quiet baseline, then score the new point.

Lecture 6 · 90 minutes

## One identifier the row supports

You map a detection to a phenomenon with a taxonomy, and the mechanism is one technique identifier per alert, backed by a field in the row. MITRE ATT&CK is the taxonomy you use for the names. You attach an identifier only when the row supports it. Four failed logins for one account in 1 minute can support a guessing label. A row that only says connection opened does not. You do not attach 12 identifiers to look thorough.

You keep tactics and techniques in their lanes. A tactic is a goal name. A technique is a narrower behavior. If your alert shows 4 failures and you only write the tactic credential access, you have not said what happened. The technique identifier is the sentence that can be checked against the 4 failures. CS 229 does not teach these names, so you read the taxonomy beside the estimator, not instead of it.

A name is allowed only with a field behind it.

1. Read the row fields

2. Pick one technique

3. Reject a bare tactic

4. Leave unsupported names off

### Four failures, one name

1. The row shows 4 failed logins for one account inside 1 minute. That field can support one guessing technique identifier.

2. You do not stop at the tactic name credential access. The technique has to be checkable against the 4 failures.

3. A different row that only says connection opened gets no identifier. You do not add 12 names. CS 229 remains the estimator course, and ATT&CK remains the name list.

Problem 1. An alert shows 4 failed logins in 1 minute, and the writeup names only the tactic credential access. What is missing?

Answer. The technique identifier that the 4 failures can check. A tactic is only a goal name. You attach 1 identifier the row supports, not 12, and a bare connection-opened row gets none.

Problem 2. You attached 12 technique identifiers to one connection-opened row so the map would look complete. How many does the row support in this example?

Answer. 0. Connection opened does not support a guessing label. Four failed logins for one account in 1 minute could support 1 identifier. CS 229 does not supply these names.

Lecture 7 · 90 minutes

# Buckets before color

Visualization for this course is a count you can recompute, and the mechanism is a time bucket. You slice one hour into 6 buckets of 10 minutes. If the failure counts in those buckets are 1, 1, 1, 1, 1, and 9, the last bucket holds 9 of the 14 failures. You plot the 6 numbers, and you do not smooth them into a single hourly total of 14 before you look. The spike is the 9.

You annotate the bucket with the rows that fell in it, or the picture cannot be audited. The 9 failures should point at 9 rows, each with the 5 fields from the first lecture. If 2 of the 9 are missing an account, you have 7 usable rows, and you say 7. Color is not a scale. If you paint high counts red, you still print the numeral 9, because a reader cannot reverse a color into a count.

From one hour to a spike you can audit.

Cut six 10-minute buckets -> Count failures in each -> Point at the rows -> Print the numeral

## Nine of fourteen in the last bucket

1. The six buckets hold 1, 1, 1, 1, 1, and 9 failures. The sum is 14, and the last bucket holds 9.

2. You plot all 6 numbers. You do not replace them with the single total 14 before you look.

3. Two of the 9 rows lack an account, so you cite 7 usable rows. The chart still prints 9, even if that bar is red.

Problem 1. Bucket counts are 1, 1, 1, 1, 1, and 9. What is the hourly total, and how many failures sit in the last bucket?

Answer. The total is 14, and the last bucket holds 9. You plot the 6 numbers. Smoothing to 14 before you look hides the spike.

Problem 2. The last bucket has 9 failure rows and 2 of them lack an account. How many usable rows do you cite?

Answer. 7. 9 minus 2 is 7. You still print the numeral 9. Color is not a scale, and each usable row should carry the 5 fields.

Lecture 8 · 90 minutes

# What you can now measure

You can now turn a log into a count someone else can repeat. A row has 5 fields, and a label is not copied from the action text. A base rate of 2 out of 50 is 0.04, and the same count out of 200 is 0.01. Precision on 2 true flags out of 3 flags is 2/3. The distance from (0, 0) to (3, 4) is 5. Mean 4 and threshold 5 flag a distance of 6.

The methods path is CS 229, with anomaly detection included, and the names path is MITRE ATT&CK. You do not swap them. After both, you practice on 1 public dataset. Your closing check is the mean of 3, 4, 4, and 5, which is 4, and the distance of the point 10, which is 6. If you cannot show that arithmetic, or you cannot say which 1 technique the 4 failures support, this course is not finished.

Counts that close the course.

Base rate before praise

Precision on held-out rows

Distance against a cutoff

One technique per alert

## Mean 4, distance 6

1. Quiet counts 3, 4, 4, and 5 sum to 16. Divided by 4, the mean is 4. The point 10 sits at distance 6, which clears a threshold of 5.

2. Separately, (0, 0) to (3, 4) is distance 5, and 2 true flags out of 3 is precision 2/3.

3. You name 1 ATT&CK technique for the 4 failures, using CS 229 only for the estimator. A missing subtraction means this course is not finished.

Problem 1. What is the mean of 3, 4, 4, and 5, and what is the distance from that mean to 10? Does a threshold of 5 flag it?

Answer. The mean is 4, because 16 divided by 4 is 4. The distance is 6, because 10 minus 4 is 6. Yes, 6 clears a threshold of 5. Folding 10 into the mean is a different, worse summary.

Problem 2. Precision is 2 true flags out of 3 flags, and the points (0, 0) and (3, 4) are one cluster candidate. What are the precision and the distance?

Answer. Precision is 2/3. The distance is 5, because 9 plus 16 is 25 and the root of 25 is 5. You still owe 1 technique name for a row that shows 4 failures, or this course is not finished.